

Supplementary Material

Adaptive and Resource-Aware Retrieval-Augmented Generation

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Scope and version

This supplement corresponds to the corpus frozen on 10 September 2026 and the 73-study manuscript version. It records the normalized search strategy, PRISMA 2020 alignment and count reconciliation, coding definitions, complete screened-corpus inventory, complete study-level appraisal, and a compact benchmarking checklist. Any later corpus update should receive a new freeze date and a new PRISMA flow.

Supplementary Table S1. Search strategy and database-specific strings

The historical screening log preserves the database-level counts reported in the main manuscript. The strings below are the normalized, reproducible formulations corresponding to the four recorded concept blocks; platform-specific field syntax and punctuation were adapted to each interface. They should therefore be read as the reproducible search logic rather than as exported UI histories.

Source	Database-specific normalized search string
ACL Anthology	("retrieval augmented generation" OR RAG OR "retrieval augmented LLM" OR "retrieve and generate") AND (adaptive OR dynamic OR active OR iterative OR routing OR triggering OR stopping OR sufficiency OR "information gap" OR controller OR policy OR uncertainty OR calibration OR budget OR latency OR "retrieval depth")
ACM Digital Library	[[All: "retrieval augmented generation"] OR [All: "retrieval augmented LLM"] OR [All: RAG]] AND [[All: adaptive] OR [All: dynamic] OR [All: iterative] OR [All: routing] OR [All: triggering] OR [All: stopping] OR [All: controller] OR [All: uncertainty] OR [All: "cost aware"] OR [All: budget] OR [All: "retrieval depth"]]
IEEE Xplore	("All Metadata":"retrieval augmented generation" OR "All Metadata":"retrieval augmented LLM" OR "All Metadata":RAG) AND ("All Metadata":adaptive OR dynamic OR iterative OR routing OR triggering OR stopping OR controller OR uncertainty OR calibration OR "resource aware" OR "cost aware" OR budget OR latency)
OpenReview	("retrieval augmented generation" OR "retrieval augmented LLM" OR RAG) AND (adaptive OR dynamic OR active OR iterative OR routing OR triggering OR stopping OR sufficiency OR controller OR policy OR uncertainty OR budget OR latency OR "retrieval depth")
SpringerLink	("retrieval augmented generation" OR "retrieval augmented LLM" OR RAG) AND (adaptive OR dynamic OR active OR iterative OR routing OR triggering OR stopping OR sufficiency OR controller OR policy OR uncertainty OR calibration OR "resource-aware" OR "cost-aware" OR budget)
ScienceDirect	("retrieval-augmented generation" OR "retrieval augmented LLM" OR RAG) AND (adaptive OR dynamic OR active OR iterative OR routing OR triggering OR stopping OR sufficiency OR controller OR policy OR uncertainty OR calibration OR "resource-aware" OR "cost-aware" OR budget OR latency)
arXiv	(all:"retrieval augmented generation" OR all:"retrieval augmented LLM" OR all:RAG) AND (all:adaptive OR all:dynamic OR all:active OR all:iterative OR all:routing OR all:triggering OR all:stopping OR all:sufficiency OR all:controller OR all:policy OR all:uncertainty OR all:budget OR all:latency)

The database search was supplemented by backward and forward citation snowballing. Records were deduplicated across preprint and peer-reviewed versions, with the peer-reviewed record retained when both referred to the same study.

Supplementary Table S2. PRISMA 2020 alignment and count reconciliation

PRISMA reporting element	Location / implementation in this review
Title / review type	Title and abstract identify the work as a systematic review of adaptive and resource-aware RAG.
Rationale and objectives	Introduction and seven review questions in Section 3.1.
Eligibility criteria	Section 3.3; inference-time adaptivity, LLM/MLLM retrieval-generation setting, venue/full-text policy, and explicit exclusions.
Information sources	Section 3.2 and Supplementary Table S1.
Search strategy	Four normalized concept blocks in main-text Table 1; database-specific normalized strings in Table S1.
Selection process	Section 3.3 and Figure 3.
Data collection / coding	Section 3.6 and Supplementary Table S3.
Study appraisal	Section 3.7 and Supplementary Table S4B.
Synthesis methods	Taxonomy, controller comparison, evaluation framework, evidence-linked failure synthesis, and research agenda in the main manuscript.
Study selection results	Figure 3 and the reconciliation below.
Included-study characteristics	Appendix A and the structured 73-study CSV.
Appraisal results	Main-text appraisal summary and Supplementary Table S4B.
Limitations	Methodology notes the absence of independent human double-coding/IRR and the full-text gate; open issues distinguish evidence gaps from method failures.
Registration / protocol	No prospective external registration was performed; the frozen search/coding artifacts and corpus version are supplied for auditability.
Data and materials	Structured corpus CSV, appraisal CSV, and this supplement accompany the source package.

Count reconciliation. Database records: $1,245 + 620 + 580 + 710 + 540 + 590 + 527 = 4,812$. Citation snowballing adds 318, giving 5,130 identified records. Removing 1,642 duplicates leaves 3,488 title/abstract records. Excluding 3,215 leaves 273 full texts. Full-text exclusions sum to $42 + 78 + 31 + 18 + 12 + 19 = 200$, leaving 73 included studies. Corpus composition is $69 + 3 + 1 = 73$ (peer-reviewed, preprint, workshop/preprint).

Supplementary Table S3. Coding and appraisal definitions

Field	Operational definition
Primary control emphasis	One descriptive code per record: Triggering; Routing; Budget/evidence; Stopping/sufficiency. Multi-phase behavior remains in the detailed taxonomy.
AD	Adaptive decision clarity: the inference-time decision/action is explicitly identifiable.
CO	Controller observability: signal/state and resulting action are sufficiently specified.
BL	Baseline adequacy: relevant fixed and/or adaptive baselines support the principal claim.
AB	Ablation quality: important controller components or decision factors are experimentally isolated.
RR	Resource reporting: relevant retrieval calls, tokens, depth, latency, memory, compute, or monetary cost are reported.
EB	Evaluation breadth: multiple datasets, tasks, domains, models, or meaningful transfer settings are evaluated.
SR	Statistical robustness: repeated runs, uncertainty intervals, significance tests, or comparable variability analysis are reported.
RP	Reproducibility: code/data/model artifacts or sufficiently complete implementation details are available.
FA	Failure/robustness analysis: failure cases, sensitivity, calibration, OOD, noisy/conflicting evidence, or related stress tests are examined.
CE	Claim–evidence alignment: principal claims remain bounded by the reported experiments and adaptive locus.
Codes	C = clearly reported/substantively addressed; P = partial/limited; NR = not clearly established in the inspected source or linked official artifact. NR is not proof that an analysis was never performed.

Reliability note. The appraisal was not independently double-coded by a second human reviewer. Consequently, Cohen’s κ or another inter-rater reliability statistic is not reported. The coding definitions and complete matrix are exposed to make this limitation auditable. A future update should independently recode a prespecified stratified subset before reporting IRR.

Supplementary Table S4A. Complete screened-corpus inventory

The inventory below preserves the record-level composition of the 73-study corpus used for the main-text synthesis. Primary control emphasis is a descriptive code; multi-phase behavior is retained in the main taxonomy.

Year	Study	Primary control emphasis	Venue	Record status
2023	IRCoT	Routing	ACL	Peer-reviewed
2023	Iter-RetGen	Routing	Findings of EMNLP	Peer-reviewed
2023	Rewrite-Retrieve-Read	Routing	EMNLP	Peer-reviewed
2023	FLARE	Triggering	EMNLP	Peer-reviewed
2023	SKR	Triggering	Findings of EMNLP	Peer-reviewed
2024	EfficientRAG	Budget/evidence	EMNLP	Peer-reviewed
2024	RE-RAG	Budget/evidence	EMNLP	Peer-reviewed
2024	REAR	Budget/evidence	EMNLP	Peer-reviewed
2024	Adaptive-RAG	Routing	NAACL	Peer-reviewed
2024	CRAG	Routing	arXiv	Preprint
2024	Self-Route	Routing	EMNLP Industry	Peer-reviewed
2024	Query-Knowledge Relevance	Stopping/sufficiency	EMNLP	Peer-reviewed
2024	Conversational Adaptive RAG	Triggering	Findings of EMNLP	Peer-reviewed
2024	DRAGIN	Triggering	ACL	Peer-reviewed
2024	RetrievalQA	Triggering	Findings of ACL	Peer-reviewed
2024	Self-RAG	Triggering	ICLR	Peer-reviewed
2024	UAR	Triggering	Findings of EMNLP	Peer-reviewed
2025	AMBER	Budget/evidence	Findings of ACL	Peer-reviewed
2025	Adaptive-k	Budget/evidence	EMNLP	Peer-reviewed
2025	Overlap Reduction	Budget/evidence	Findings of ACL	Peer-reviewed
2025	DeepRAG	Routing	arXiv	Preprint
2025	MBA-RAG	Routing	COLING	Peer-reviewed
2025	Q-DREAM	Routing	ACL	Peer-reviewed
2025	RAG-Gym	Routing	arXiv	Preprint
2025	Search-R1	Routing	COLM	Peer-reviewed
2025	Search-o1	Routing	EMNLP	Peer-reviewed
2025	Smart-Searcher	Routing	Findings of EMNLP	Peer-reviewed
2025	UniRAG	Routing	ACL	Peer-reviewed
2025	Sufficient Context	Stopping/sufficiency	ICLR	Peer-reviewed
2025	Adaptive Retrieval Comparison	Triggering	ACL	Peer-reviewed
2025	CtrlA	Triggering	Findings of ACL	Peer-reviewed
2025	DioR	Triggering	ACL	Peer-reviewed
2025	LLM-Independent Adaptive RAG	Triggering	EMNLP	Peer-reviewed
2025	SeaKR	Triggering	ACL	Peer-reviewed
2025	EI-ARAG	Triggering	COLING	Peer-reviewed
2025	SGIC	Budget/evidence	ACL	Peer-reviewed
2026	BRIEF-Pro	Budget/evidence	Findings of ACL	Peer-reviewed

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Supplementary Table S4A continued.

Year	Study	Primary control emphasis	Venue	Record status
2026	Flow-RAG	Budget/evidence	Knowledge-Based Systems	Peer-reviewed
2026	IterCOMP	Budget/evidence	ACL	Peer-reviewed
2026	RAM	Budget/evidence	ACL	Peer-reviewed
2026	SARA	Budget/evidence	ACL	Peer-reviewed
2026	AIR-RAG	Routing	Neurocomputing	Peer-reviewed
2026	Agentic-R	Routing	Findings of ACL	Peer-reviewed
2026	CIRAG	Routing	ACL	Peer-reviewed
2026	DORA	Routing	ACL	Peer-reviewed
2026	HiKEY	Routing	ACL	Peer-reviewed
2026	PROGRAM	Routing	Findings of ACL	Peer-reviewed
2026	QuDAR	Routing	ACL	Peer-reviewed
2026	RouteRAG	Routing	Findings of ACL	Peer-reviewed
2026	SPARKLE	Routing	ACL	Peer-reviewed
2026	AdaKG-RAG	Stopping/sufficiency	Journal of King Saud University Computer and Information Sciences	Peer-reviewed
2026	AutoSearch	Stopping/sufficiency	Findings of ACL	Peer-reviewed
2026	Reflective RAG	Stopping/sufficiency	Findings of ACL	Peer-reviewed
2026	ReflectiveRAG	Stopping/sufficiency	EACL Industry	Peer-reviewed
2026	S2G-RAG	Stopping/sufficiency	ACL	Peer-reviewed
2026	Budget-Aware Active RAG Evaluation	Triggering	KDD 2026 Workshop / arXiv	Workshop/preprint
2026	INKER	Triggering	Information Processing & Management	Peer-reviewed
2026	D2-RAG	Triggering	Findings of ACL	Peer-reviewed
2026	AED-RAG	Budget/evidence	Findings of ACL	Peer-reviewed
2026	GRIP	Routing	ACL	Peer-reviewed
2026	ReaLM-Retrieve	Triggering	SIGIR	Peer-reviewed
2026	Cost-aware depth	Budget/evidence	The Web Conference	Peer-reviewed
2026	PruneRAG	Stopping/sufficiency	The Web Conference	Peer-reviewed
2026	LogicRAG	Routing	AAAI	Peer-reviewed
2026	DA-RAG	Budget/evidence	The Web Conference	Peer-reviewed
2026	TAdARAG	Routing	AAAI	Peer-reviewed
2026	CompSelect	Budget/evidence	The Web Conference	Peer-reviewed
2026	GRAD	Budget/evidence	ACL	Peer-reviewed
2026	REPAIR	Routing	ACL	Peer-reviewed
2026	RAG-on-a-Diet	Budget/evidence	ACL	Peer-reviewed
2025	FISKE	Stopping/sufficiency	Knowledge-Based Systems	Peer-reviewed
2026	QR ³ AG	Triggering	Expert Systems with Applications	Peer-reviewed
2026	SC-RAG	Budget/evidence	Information Processing & Management	Peer-reviewed

Supplementary Table S4B. Complete study-level appraisal

Year	Study	AD	CO	BL	AB	RR	EB	SR	RP	FA	CE
2023	FLARE	C	C	C	C	C	C	NR	C	P	C
2023	IRCoT	C	C	C	P	C	C	NR	C	P	C
2023	Iter-RetGen	C	C	C	P	P	C	NR	P	NR	C
2023	Rewrite-Retrieve-Read	C	C	C	P	P	C	NR	P	NR	C
2023	SKR	C	C	C	P	P	C	NR	P	P	C
2024	Adaptive-RAG	C	C	C	C	C	C	P	C	P	C
2024	CRAG	C	C	P	P	P	C	NR	NR	P	C
2024	Conversational Adaptive RAG	C	C	C	P	P	C	NR	P	P	C
2024	DRAGIN	C	C	C	C	C	C	NR	C	C	C
2024	EfficientRAG	C	C	C	C	C	C	NR	C	P	C
2024	Query-Knowledge Relevance	C	C	P	NR	NR	P	NR	P	P	C
2024	RE-RAG	C	C	C	C	P	C	NR	P	C	C
2024	REAR	C	P	C	C	NR	C	NR	P	C	C
2024	RetrievalQA	C	C	C	P	P	C	NR	P	NR	C
2024	Self-RAG	C	C	C	C	C	C	P	C	C	C
2024	Self-Route	C	C	C	P	P	C	NR	P	NR	C
2024	UAR	C	C	C	P	P	C	NR	P	P	C
2025	AMBER	C	P	C	P	P	C	NR	P	P	P
2025	Adaptive Retrieval Comparison	C	C	P	NR	NR	C	NR	P	C	C
2025	Adaptive-k	C	C	C	C	C	C	NR	P	P	C
2025	CtrlA	C	C	C	P	P	C	NR	P	P	C
2025	DeepRAG	C	C	P	P	C	C	NR	NR	P	C
2025	DioR	C	C	C	C	P	C	NR	P	C	C
2025	EI-ARAG	C	C	C	C	C	C	NR	C	P	C
2025	FISKE	C	C	C	C	C	C	NR	C	C	C
2025	LLM-Independent Adaptive RAG	C	C	C	P	P	C	NR	P	P	C
2025	MBA-RAG	C	C	C	P	C	C	NR	P	P	C
2025	Overlap Reduction	C	P	C	P	C	P	NR	P	NR	P
2025	Q-DREAM	C	P	C	C	P	C	P	C	C	P
2025	RAG-Gym	C	C	P	NR	C	C	NR	P	P	C
2025	SGIC	C	P	C	P	P	P	NR	P	P	C
2025	SeaKR	C	C	C	C	C	C	P	C	C	C
2025	Search-R1	C	C	C	C	C	C	P	C	P	C
2025	Search-o1	C	C	C	C	C	C	NR	C	P	C
2025	Smart-Searcher	C	C	C	P	C	C	NR	P	NR	C
2025	Sufficient Context	C	C	P	NR	NR	C	NR	P	C	C
2025	UniRAG	C	C	C	P	P	C	NR	P	P	C
2026	AED-RAG	C	C	C	C	P	C	NR	P	C	C
2026	AIR-RAG	C	C	C	C	P	C	P	P	C	C
2026	AdaKG-RAG	C	C	C	C	P	P	P	P	C	C
2026	Agentic-R	C	C	C	C	C	C	P	C	C	C
2026	AutoSearch	C	C	C	C	C	C	P	C	C	C

Year	Study	AD	CO	BL	AB	RR	EB	SR	RP	FA	CE
2026	BRIEF-Pro	C	P	C	P	C	P	NR	P	NR	P
2026	Budget-Aware Active RAG Evaluation	C	C	P	NR	C	P	NR	NR	P	P
2026	CIRAG	C	C	C	C	C	C	NR	P	C	C
2026	CompSelect	C	C	C	C	C	C	P	C	C	C
2026	Cost-aware depth	C	C	C	C	C	C	C	P	C	C
2026	D2-RAG	C	C	C	C	P	C	P	P	C	C
2026	DA-RAG	C	C	C	C	C	P	P	C	C	C
2026	DORA	C	C	C	C	C	C	P	C	C	C
2026	Flow-RAG	C	C	C	C	C	P	P	C	C	C
2026	GRAD	C	P	C	C	NR	C	NR	C	NR	P
2026	GRIP	C	C	C	C	C	C	P	C	P	C
2026	HiKEY	C	C	C	C	C	P	NR	P	P	C
2026	INKER	C	C	C	C	C	C	P	P	C	C
2026	IterCOMP	C	C	C	P	C	P	NR	P	P	C
2026	LogicRAG	C	C	C	C	C	C	NR	C	P	C
2026	PROGRAM	C	C	C	P	C	C	NR	P	P	C
2026	PruneRAG	C	C	C	C	C	C	P	C	C	C
2026	QR3AG	C	C	C	C	C	C	NR	P	P	C
2026	QuDAR	C	C	C	C	P	C	P	C	C	C
2026	RAG-on-a-Diet	C	C	C	C	C	C	C	C	C	C
2026	RAM	C	C	C	P	C	P	NR	P	P	C
2026	REPAIR	C	C	C	C	C	C	P	C	C	C
2026	ReaLM-Retrieve	C	C	C	C	C	C	P	C	C	C
2026	Reflective RAG	C	C	C	P	P	C	NR	P	P	C
2026	ReflectiveRAG	C	C	C	P	P	C	NR	P	P	C
2026	RouteRAG	C	C	C	C	C	C	NR	P	P	C
2026	S2G-RAG	C	C	C	C	C	C	P	C	C	C
2026	SARA	C	C	C	P	C	P	NR	P	P	C
2026	SC-RAG	C	P	C	C	C	C	NR	P	C	C
2026	SPARKLE	C	C	C	C	C	C	P	C	C	C
2026	TAdaRAG	C	P	C	C	P	C	NR	P	P	P

Clear-reporting rates by primary control locus (%). This stratified view is diagnostic rather than a maturity ranking; it shows where reporting is comparatively complete or sparse without treating the ten criteria as

	Locus	<i>n</i>	AD	CO	BL	AB	RR	EB	SR	RP	FA	CE
interchangeable.	Triggering	18	100.0	100.0	88.9	55.6	50.0	94.4	0.0	33.3	44.4	94.4
	Routing	27	100.0	92.6	88.9	59.3	66.7	96.3	0.0	44.4	29.6	92.6
	Budget/evidence	19	100.0	63.2	100.0	63.2	68.4	57.9	10.5	31.6	47.4	78.9
	Stopping/sufficiency	9	100.0	100.0	77.8	55.6	44.4	77.8	0.0	44.4	66.7	100.0

Supplementary Table S5. Benchmarking checklist

Layer	Recommended minimum reporting
Answer quality	Report final task quality against fixed-retrieval and, where meaningful, no-retrieval base-lines.
Retrieval quality	Report evidence recall/precision, supporting-fact/path coverage, or an appropriate retrieval metric when gold evidence exists.
Decision quality	Measure the error type corresponding to the adaptive locus: false retrieve/skip, routing error, sufficiency error, premature/late stopping, or abstention error.
Retrieval utility	Estimate whether the chosen retrieval/evidence action improved the final answer, including harm or marginal-utility analysis where feasible.
Resource efficiency	Report controller overhead and realized retrieval calls, documents processed, context tokens, search depth, LLM/tool calls, latency, memory, and monetary/compute cost when relevant.
Robustness	Repeat or stress-test key quality-cost claims under threshold changes, model/retriever changes, and irrelevant/conflicting evidence where applicable.
Reproducibility	Record corpus snapshot, chunking, retriever/index version, model version, controller settings, seeds, and public traces/artifacts when possible.